

How Interconnection Shapes Synchronization in Oscillator Networks

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MOTIVATION

When integrating two independently operating power networks, the interconnection is rarely treated as a deliberate design choice – yet it directly governs how phase perturbations propagate across the interlink and how quickly the merged system recovers synchronization. Prolonged frequency deviations following integration can cascade into system-wide instability, making the selection of bridging nodes a critical but underexplored factor in network merging strategies.

Idea: Connecting networks through their center-of-mass nodes stabilizes the merged system more effectively than through purely topological centers.

METHOD

Swing Equation Dynamics

We generate **two independent SHK networks without self-loops** and will merge them via inter-network links (strategies followed). The coupled-oscillator dynamics is governed by the second-order Kuramoto model. Each node i carries a binary property value $P_i \in \{+1, -1\}$ representing a power producer or consumer.

Network sizes $N_1 = 100, N_2 = 200 \rightarrow N_{\text{merged}} = 300$

SHK parameters $p = 0.01, q = 0.32, r = 1, s = 0$

Ensemble 30 independent realizations

$$\dot{\omega}_i = P_i - \alpha \cdot \omega - \sum_j K_{ij} \sin(\theta_j - \theta_i) \quad d\theta_i/dt = \omega_i$$

θ_i : Angle of node i

ω_i : Angular velocity of node i

P_i : Power production or consumption

α : Damping Coefficient

K_{ij} : Transmission Capacity

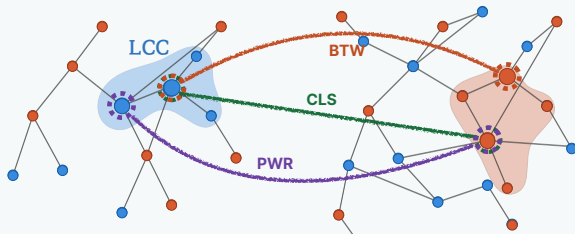
MERGE STRATEGIES & STABILITY METRICS

BTW — Connect the highest **betweenness centrality** node in the Largest Connected Component (LCC) of each group.

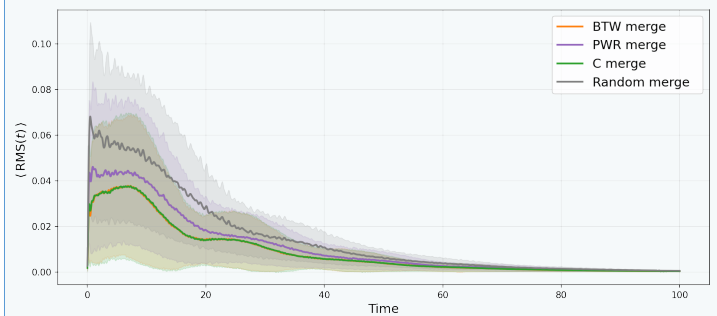
CLS — Connects the node with the highest **closeness centrality** in the LCC of each group.

PWR — Connect the node **closest** to the **power-weighted center** of mass $\left(\frac{\sum |P_i| x_i}{\sum |P_i|}, \frac{\sum |P_i| y_i}{\sum |P_i|} \right)$ within the LCC of each group.

● Producer ● Consumer



RESULTS — RMS FLUCTUATION



$$\langle \text{RMS}(\omega(t)) \rangle: \left\langle \sqrt{\frac{1}{N} \sum_i \omega_i(t)^2} \right\rangle, \quad F_{\text{rms}}: \left\langle \sqrt{\frac{1}{T} \sum_t \text{RMS}(t)^2} \right\rangle$$

Strategy	F_{rms}	Fluctuation
BTW	0.01405	60.1%
CLS	0.01409	60.3%
PWR	0.01750	74.9%
Random	0.02335	Baseline

BTW (0.01405) and CLS (0.01409) show approximately 40% lower fluctuation than the random baseline, despite using only a single inter-network edge. PWR (0.01750) also improves over random but falls short of the shortcut-centrality-based strategies.

RESULTS — RECOVERY TIME

Mean T_{rec} : the mean of converged time.

Strategy	Mean T_{rec}	Improvement
CLS	15.506	48.3%
BTW	16.563	44.8%
PWR	21.118	29.6%
Random	29.988	Baseline

All three strategies outperform random merging, with CLS (48.3%) and BTW (44.8%) achieving nearly half the recovery time of the baseline, while PWR (29.6%) shows a more modest improvement – consistent with its insensitivity to node power values.

ONGOING WORK & FUTURE DIRECTIONS

↑ **Why centrality-based merging works** : High betweenness/closeness nodes sit on many shortest paths within their cluster, enabling rapid phase coordination

↓ **Why PWR underperform**: $|P_i|$ cancels out in the centroid formula, making node power values irrelevant to node selection. The resulting node is not necessarily a topological hub, slowing inter-cluster phase propagation.

→ **Refining PWR**: Redefine the center of mass to genuinely reflect power flow heterogeneity – e.g., weighting by $|P_i|$ or local flow capacity so the selected node is central in the power-flow sense, not just geometry.

→ **Realistic Power distribution and topology**: Extend to network models with greater heterogeneity in P_i sizes and validate on real power grid topologies to test the robustness of centrality-based merging under realistic conditions.